

MEASURING THE MOAT: THE MINIMUM EXCESS OF SEYMOUR ORIENTATIONS WITH LARGE MINIMUM OUT-DEGREE

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ABSTRACT. Seymour’s second neighbourhood conjecture (SSNC) asserts that every oriented graph has a vertex whose second out-neighbourhood is at least as large as its first. A counterexample would be an oriented graph in which *every* vertex falls short. We study how close one can get. For an oriented graph G define the **excess** $\text{exc}(G) = \sum_v \max(0, d_2(v) - d_1(v) + 1)$, where $d_i(v) = |N_i^+(v)|$; thus $\text{exc}(G) = 0$ iff G is a counterexample to SSNC, and SSNC is equivalent to $\text{exc}(G) \geq 1$ for all G . Since excess 1 is trivially attained using sinks, we impose the minimum out-degree bound $\delta^+ \geq 8$ forced on counterexamples by Kaneko–Locke and the recent $\delta^+ = 7$ elimination, and define $E(n) = \min\{\text{exc}(G)\}$ over strongly connected oriented graphs on n vertices with $\delta^+ \geq 8$.

We give constructive upper bounds ($E(n) \leq m$ whenever $n = mt$, $m \geq 3$, $t \geq 8$, with a generalization to cycle-power skeletons covering many further n), and a transplant construction derived from a stochastic-search minimizer that yields $E(n) \leq 5$ at several primes ($n = 47, 53, 59$). We report machine-verified measurements for $24 \leq n \leq 75$, all witnesses published with hashes and re-verifiable by three independent implementations. On the lower-bound side we prove $E(17) \geq 3$ by constraint-solver infeasibility, and report that the measured floor at $n = 50$ is invariant under raising the degree bound to $\delta^+ \geq 9, 10$. Structural analysis of minimizers reveals two distinct local architectures (a distributed “ring membrane” and a concentrated “keystone cluster”) which nevertheless share identical local dynamics: strict single-flip local minimality, large neutral plateaus, and stability under all $\sim 1.7 \times 10^5$ coordinated two-flip attacks tested. We conjecture a uniform lower bound $E(n) \geq 3$ and pose the regularity of $n \mapsto E(n)$ as an open problem. All claims are reproducible from the accompanying repository.

1. INTRODUCTION

Seymour’s second neighbourhood conjecture [DL95] states:

Conjecture 1.1 (SSNC). *Every oriented graph (digraph without 2-cycles) contains a vertex v with $|N_2^+(v)| \geq |N_1^+(v)|$.*

The conjecture is proved for tournaments [Fis96, HT00] and in many special classes; every vertex satisfies $|N_2^+(v)| \geq \gamma |N_1^+(v)|$ for some vertex

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with $\gamma = 0.715538\dots$ [HP24], improving [CSY03]. A counterexample must have minimum out-degree $\delta^+ \geq 8$: out-degree ≤ 6 was eliminated by [KL01] and $\delta^+ = 7$ by [SSS26] via a signature-compression argument checked by CP-SAT.

Recent work has begun to study the *equality boundary* of the conjecture. [Hal26] calls a strongly connected oriented graph **Pisa** if $\max_v(|N_2^+(v)| - |N_1^+(v)|) = 0$, i.e. no vertex exceeds and at least one vertex meets the bound; [GKZ26] study **Seymour-tight** orientations, in which *every* vertex meets the bound, and develop a product calculus and an algebraic classification for them. In the tight case the set of boundary vertices is everything; in a generic Pisa graph it can be much smaller. This note measures how small.

Definitions. For an oriented graph G write $d_1(v) = |N_1^+(v)|$, $d_2(v) = |N_2^+(v)|$, and call $m(v) = d_2(v) - d_1(v)$ the *margin* of v . Call v **tight** if $m(v) = 0$ and **satisfied** if $m(v) \geq 0$ (a “Seymour vertex”). Define

$$\text{exc}(G) = \sum_{v \in V(G)} \max(0, m(v) + 1).$$

Then $\text{exc}(G) = 0$ iff G is a counterexample to SSNC; SSNC is the statement $\text{exc} \geq 1$; and for a Pisa graph the excess equals the number of tight vertices. Adding a universal sink produces $\text{exc} = 1$ trivially [GKZ26, §1], so the interesting quantity requires the degree bound that counterexamples must satisfy:

$$E(n) = \min\{\text{exc}(G) : |V(G)| = n, G \text{ strongly connected oriented}, \delta^+(G) \geq 8\}.$$

(Strong connectivity is justified by the standard minimal-counterexample reduction: a closed out-section of a counterexample is a smaller counterexample; cf. [EDGGK25].)

$E(n) = 0$ for some n iff SSNC is false. If SSNC is true, then $E(n) \geq 1$ for all n , and any stronger uniform lower bound is a strengthening of the conjecture. The contributions of this note:

- (1) **Upper bounds by construction** (§2): a single “out-set inclusion” lemma yields both a family of *pure-ring* constructions ($E(mt) \leq m$ for $m \geq 3$, $t \geq 8$; generalized to cycle-power skeletons) and, after transplant of a search-discovered local structure, the bound $E(n) \leq 5$ at the primes $n = 47, 53, 59$.
- (2) **Measurements** (§3): a table of best known upper bounds for $24 \leq n \leq 75$ with published, hash-identified witnesses; and a measurement-theoretic caveat we learned the hard way — *search attainment is not the floor* — documented via a controlled experiment.
- (3) **Lower bounds and rigidity** (§4): $E(17) \geq 3$ (CP-SAT infeasibility); invariance of the measured $n = 50$ floor under $\delta^+ \geq 9, 10$; and twelve one-hour UNKNOWNs for the tournament-restricted question $\text{exc} \leq 2$, $17 \leq n \leq 28$.
- (4) **Structure of minimizers** (§5): two architectures, one dynamics.

- (5) **Open problems** (§6), foremost: does a uniform constant lower bound $E(n) \geq c \geq 2$ hold, and what governs the apparent floor $3 \leq E(n) \leq 5$?

Verification protocol. Every witness in this note is stored in the accompanying repository as an adjacency list with a SHA-1 fingerprint, and every numerical claim is checked by three implementations written independently (a numpy pipeline used by the search; a pure-Python verifier; and a standard-library-only verifier written from the definitions without access to the other two). A `make verify-*` target reproduces each claim.

2. UPPER BOUNDS BY CONSTRUCTION

2.1. The out-set inclusion lemma. All constructions below are perturbed blow-ups of directed cycles. The mechanism that keeps perturbations from leaking is the following observation.

Lemma 2.1 (out-set inclusion). *Let G be an oriented graph and let $v \neq p$ be non-adjacent vertices with $N_1^+(p) \subseteq N_1^+(v)$. Let $G' = G + (v \rightarrow p)$. Then G' is oriented, and:*

- (i) *if $p \in N_2^+(G, v)$, then $m_{G'}(v) = m_G(v) - 2$;*
- (ii) *if $p \notin N_1^+(G, v) \cup N_2^+(G, v)$, then $m_{G'}(v) = m_G(v) - 1$;*
- (iii) *for $x \neq v$, margins change only by the possible appearance of p in $N_2^+(x)$ through the new path $x \rightarrow v \rightarrow p$: precisely, $m_{G'}(x) = m_G(x) + 1$ exactly when $v \in N_1^+(x)$ and $p \notin N_1^+(x) \cup \{x\} \cup N_2^+(G, x)$; all other margins are unchanged. In particular, if every in-neighbour x of v satisfies $p \in N_1^+(x)$, then no margin other than $m(v)$ changes.*

Proof. G' is oriented since $p \rightarrow v$ is absent by non-adjacency. For v : $d_1(v)$ increases by exactly one. Candidate new second neighbours of v are the out-neighbours of the new first neighbour p , i.e. $N_1^+(p) \setminus (N_1^+(G', v) \cup \{v\}) = \emptyset$ by hypothesis; and no old second neighbour $u \neq p$ is lost, since arcs are only added and $u \notin N_1^+(G', v)$. The only possible change to $N_2^+(v)$ is the removal of p itself, which occurs precisely in case (i); this gives (i) and (ii). For $x \notin \{v, p\}$: $d_1(x)$ is unchanged, and the only new directed paths of length two are of the form $x \rightarrow v \rightarrow p$, which affect membership of p alone; the stated condition characterizes when p actually enters $N_2^+(x)$. For $x = p$: no new path leaves p , and no new path of length two into $N_2^+(p)$ is created, since such a path would require an arc into v to be new. $\square$

In words: *if the target's out-set is already inside the shooter's first neighbourhood, the shot costs the shooter one unit of margin and is invisible to bystanders who see both.* The same arithmetic underlies the in-star trick in the generalized lexicographic products of [GKZ26] — and, as §5 shows, it was re-invented independently by stochastic search.

2.2. Pure rings.

Theorem 2.2. *Let $m \geq 3$ and $t \geq 8$, and let $n = mt$. Then $E(n) \leq m$.*

Construction. Partition V into layers $L_0, \dots, L_{m-1}$ of size t ; add all arcs $L_i \rightarrow L_{i+1}$ (indices mod m); in each layer choose one *pure* vertex p_i and add the arc $v \rightarrow p_i$ for every *impure* $v \in L_i \setminus \{p_i\}$. Every impure vertex has $d_1 = t + 1$, $d_2 = t$, hence margin -1 ; every pure vertex has $d_1 = d_2 = t$, margin 0 . The graph is digon-free (arcs between layers all point forward; inside a layer only $\rightarrow p_i$ arcs exist and p_i sends none), strongly connected, with $\delta^+ = t \geq 8$, and $\text{exc} = m$. In the layered construction the target p_i lies in the shooter's own layer, at directed distance $m \geq 3$ from every impure $v \in L_i$, so case (ii) of Lemma 2.1 applies and each intra-layer arc costs its shooter exactly one unit of margin; every in-neighbour of v (all of L_{i-1}) sees p_i in its first neighbourhood, so by (iii) no other margin moves. $\square$

Theorem 2.3 (cycle-power skeletons). *Let m, k with $2k < m$, and $t \geq 1$ with $kt \geq 8$; let $n = mt$. Then $E(n) \leq m$.*

Construction. Replace the cycle skeleton by the k -th power $\vec{C}_m^k$: each layer sends complete arcs to the next k layers. Purity and the in-star are as before; the out-set inclusion is now even cleaner, since the target's out-set lies entirely inside the shooter's first neighbourhood. In the skeleton $\vec{C}_m^k$ the shortest directed path from an impure $v \in L_i$ back to its own layer has length at least $\lceil m/k \rceil \geq 3$ (since $m > 2k$), so case (ii) of Lemma 2.1 applies exactly as before; the in-neighbour condition of (iii) holds as every layer L_{i-j} ($1 \leq j \leq k$) sees p_i in its first neighbourhood. This covers e.g. $n = 25$ ($m=5, k=2, t=5$), $n = 35$ ($5, 2, 7$), $n = 49$ ($7, 2, 7$). $\square$

Uneven layer sizes generally fail; the algebraic reason is the kernel condition for generalized lexicographic products identified in [GKZ26, Thm 4.6], which for the plain cycle forces equal sizes. Our constructions are *not* Seymour-tight — they sit at the opposite extreme of the Pisa class, with as few tight vertices as the skeleton allows.

2.3. Keystone transplants: upper bounds at primes. Ring constructions require $m \mid n$, and no admissible skeleton exists at e.g. $n = 47, 53, 59$. Stochastic search (§3) found a minimizer at $n = 53$ with a different local architecture: a **keystone cluster** — a satisfied vertex u of margin $+1$ together with *wing* vertices w whose out-sets are exact copies of $N_1^+(u)$ plus the single arc $w \rightarrow u$ (verified equality of out-sets in the witness; see `data/cluster_structure_n53.txt`). This is the out-set inclusion mechanism again, asymmetric and ring-free.

Treating the cluster as an invariant module and deterministically deleting (resp. cloning) bulk vertices yields witnesses establishing

$$E(47) \leq 5, \quad E(53) \leq 5, \quad E(59) \leq 5,$$

each verified by the three-implementation protocol, and $\text{exc} = 4$ at $n = 57$ with $no + 1$ vertex (all four survivors tight) — showing the $+1$ “insurance” is not intrinsic to the family (at $n = 57$ the ring bound $E \leq 3$ is anyway better).

3. MEASUREMENTS

3.1. Table of best known upper bounds. The full table is machine-generated from the repository data (`note/make_table.py`); hashes are canonical-adjacency SHA-1 prefixes from the witness manifest.

n	construction	search	best	witness
24	3 (ring $m=3,k=1,t=8$)	-	3	2517130b
25	5 (ring $m=5,k=2,t=5$)	12	5	93072b44
27	3 (ring $m=3,k=1,t=9$)	-	3	5b5ae497
30	3 (ring $m=3,k=1,t=10$)	3	3	(<i>thm</i>)
35	5 (ring $m=5,k=2,t=7$)	7	5	5a88d73d
40	4 (ring $m=4,k=1,t=10$)	23	4	(<i>thm</i>)
45	3 (ring $m=3,k=1,t=15$)	21	3	(<i>thm</i>)
47	5 (keystone transplant)	9	5	21f36278
48	3 (ring $m=3,k=1,t=16$)	13 (seed-free)	3	d43ca1db
49	7 (ring $m=7,k=2,t=7$)	6	6	ead3c3c3
50	5 (ring $m=5,k=1,t=10$)	5 (20 seed-free)	5	17cd7dc8
51	3 (ring $m=3,k=1,t=17$)	-	3	e5d2d5eb
53	5 (keystone transplant)	5	5	dc922e89
57	3 (ring $m=3,k=1,t=19$)	-	3	8210dcab
59	5 (keystone transplant)	7	5	3eedc493
60	3 (ring $m=3,k=1,t=20$)	33	3	(<i>thm</i>)
75	3 (ring $m=3,k=1,t=25$)	3	3	(<i>thm</i>)

Search values marked “seed-free” are the negative controls of §3.2. At $n = 57$ the transplant family also realises $\text{exc} = 4$ (all survivors tight), which is not a bound record there.

3.2. Search attainment is not the floor. We initially read fresh-search values at primes (47:9, 59:7, 53:5) as a “prime staircase”. A pre-registered controlled experiment (fresh search on the *composites* $n = 48, 50$ with all construction seeds withheld) returned 13 and 20 — far above the construction bounds 3 and 5 at the same n . Conclusion: **fresh-search values measure search reachability, not the floor**; the staircase was a measurement artefact, later confirmed when the keystone transplant put all three primes at 5. We therefore report construction bounds and search attainments in separate columns throughout. (Full logs, pre-registration commits, and the negative-control seed runs are in the repository.)

3.3. δ -invariance. At $n = 50$ the measured floor 5 is unchanged when the degree bound is raised to $\delta^+ \geq 9$ and $\delta^+ \geq 10$ (the divisor-ring witness has $\delta^+ = 10$ and satisfies all three constraints; forty generations of search under each constraint produced no improvement). Any conjectured lower

bound may thus plausibly be sought in a form not depending on δ beyond the counterexample threshold.

4. LOWER BOUNDS AND RIGIDITY

Theorem 4.1. $E(17) \geq 3$.

Proof. CP-SAT infeasibility of the constraint system “ $n = 17$, oriented, strongly connected, $\delta^+ \geq 8$, $\text{exc} \leq 2$ ”, with exact two-way reification of the distance-two indicators; fixed seed; model and log published. The model is `experiments/delta8/excess2_search.py (--n 17 --cap 2, random_seed 1, 8 workers)`, returning INFEASIBLE in 357.0 s under OR-Tools 9.15.6755; the machine-readable record, including the model hash (1d22eb15...), is `data/excess2_results.jsonl`. $\square$

For $18 \leq n \leq 22$ the same computation is UNKNOWN within 3600 s each; the direct method’s reach ends near $n = 17$.

Tournament rigidity. By [HT00], sink-free tournaments have at least two satisfied vertices; whether *exactly two*, *both tight* is attainable in a tournament (which would give $\text{exc} = 2$ and refute any $E \geq 3$ -type bound in the tournament-dense regime) appears to be open. Our CP-SAT scans of the tournament-restricted question $\text{exc} \leq 2$ for $17 \leq n \leq 28$ returned twelve consecutive UNKNOWNs at one hour each: the question is not cheaply decidable in either direction. We record this as an explicit open problem (§6).

5. STRUCTURE OF MINIMIZERS: TWO ARCHITECTURES, ONE DYNAMICS

Two verified minimizers at $\text{exc} = 5$:

- $n = 50$ (17cd7dc8): five tight survivors forming a **pentagon ring** $8 \rightarrow 16 \rightarrow 23 \rightarrow 36 \rightarrow 47 \rightarrow 8$ in which each survivor’s first neighbourhood contains the next survivor and its second neighbourhood the one after; the sensitivity matrix (all valid single arc-flips) is diagonal; for four of the five survivors the cheapest kill is an exact equal trade ($\Delta E = 0$, a reserve vertex rising from margin -1), while the fifth (v_{36}) costs $\Delta E = +2$.
- $n = 53$ (dc922e89): a **keystone cluster** — margins $(+1, 0, 0, 0)$ on $\{18; 34, 35, 36\}$ with the wings’ out-sets equal to the keystone’s; the sensitivity matrix shows the only non-diagonal row at the keystone: killing it raises both wings.

Different architecture and different economy (the ring witness’s deficits are shallow (margin histogram $\{-5:1, -4:1, -3:4, -2:17, -1:22\}$), while the keystone witness carries a deep deficit tail reaching -15) — yet identical dynamics: both are strict single-flip local minima with large neutral plateaus, and the keystone’s apparent $+1$ slack survives an exhaustive two-flip attack (171,288 coordinated pairs around the cluster; no decrease; 7,712

neutral pairs). Whatever forbids $\text{exc} < 5$ at these n is insensitive to the architecture, the economy, and (by §3.3) the degree bound — which suggests the true obstruction is a global counting phenomenon rather than a local structural one.

Remark 5.1 (Pisa structure). All minimizers above are Pisa graphs whose underlying undirected graphs are irregular and far from K_n -minus-a-matching; together with small blow-ups of directed cycles (e.g. the 2-blow-up of $\vec{C}_4$ on 8 vertices) they show that the classification of underlying graphs of Pisa graphs is substantially richer than conjectured in [Hal26, Conj. 5.1]. We have communicated explicit counterexamples to the author (private communication, July 2026).

6. OPEN PROBLEMS

- (1) **Uniform floor.** Is there $c \geq 2$ with $E(n) \geq c$ for all n ? All evidence above is consistent with $E(n) \geq 3$; we know only $E(17) \geq 3$ and, conditionally on SSNC, $E(n) \geq 1$.
- (2) **Regularity of E .** The best known upper bounds lie in $\{3, 4, 5, 6\}$ for all measured $24 \leq n \leq 75$, with the value governed by available skeletons, not by primality. Is E eventually constant? Monotone in any sense?
- (3) **Tournaments with exactly two tight vertices.** Does a sink-free tournament with exactly two satisfied vertices, both tight, exist? (Twelve one-hour CP-SAT attempts, $17 \leq n \leq 28$, all UNKNOWN.)
- (4) **Keystone family limits.** The transplant family attains 4 (at $n = 57$) with all survivors tight; is 3 attainable ring-free? Is the family's native floor below the ring floor anywhere?
- (5) **Global counting.** Find a double-counting / matrix-analytic proof of any bound $E(n) \geq 2$. The matrix S_D of [GKZ26] expresses margins as row sums; excess is the positive part of $S_D \mathbf{1}$. The spectral theory of S_D (untouched in the literature) and LP duality over flow relaxations both suggest themselves.

7. METHODS AND REPRODUCIBILITY

Search: island-model program evolution (LLM mutation operators acting on construction programs), with every discovered improvement persisted to disk at discovery time under a SHA-1 fingerprint. Verification: three independent implementations (§1); every claim maps to a `make verify-*` target. Pre-registration: search predictions were committed to the repository before the corresponding runs (commit hashes in `docs/`). Division of labour: direction and decisions by the human author; supervision, literature, and independent verification by Claude (Fable); implementation, search, and solver engineering by Claude (Code). The mathematics should be judged on the verifiability of the claims, not on the nature of the authors.

Total LLM API cost of all search and verification reported here: USD 65.54.

Repository: [URL on publication; private at draft time].

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